

variance test. This test is beyond the scope of this reference book, but readers interested in the concept can reference Mansfield (1994) or Newmark (1988).

Median Test

The median test is used to determine whether or not the medians of two or more independent samples are equal. If the medians of the two samples are statistically different from one another then the algorithms are not equivalent. This test is as follows:

Hypothesis:

- H_0 : Same medians
- H_1 : Different medians

Calculation Process:

1. Use arrival cost as the performance measure.
2. Choose two algorithms that are similar (e.g., arrival, IS, etc.). This experiment can be repeated to compare different algorithms.
3. Use a large enough number of orders and data points in each algorithm so that each has a representative sample size. Make sure that the orders traded in each algorithm are similar: size, volatility, market cap, buy/sell, and in similar market conditions.
4. Let X = set of observations from algorithm A.
Let Y = set of observations from algorithm B.
5. Determine the overall median across all the data points.
6. For each sample count the number of outcomes that are less than or equal (“LE”) to the median and the number of outcomes that are greater than (“GT”) the median. Use the table below to tally these results.

	Sample A	Sample B	Subtotal
LE overall median	a	b	(a + b)
GT overall median	c	d	(c + d)
Subtotal	(a + c)	(b + d)	(a + b + c + d) = n

7. Compute the expected frequency for each cell

$$ef_{ij} = \frac{\text{total observations in row } i + \text{total observations in column } j}{\text{overall total number of observations}}$$